

# Rows Columns and Data Types

Project: Transparent Book Recommender

## 1. Your mission

### Your learning target

I can specify a table schema.

I can check types and units as well as names.

I can distinguish structural validity from truth.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 7: Rows, Columns, and Data Types

Structured tables preserve meaning through named fields and consistent types.



## THINKER MISSION

Repair a mixed-up class-pet dataset.

Thinker name and date:

**Write or draw your reasoning and evidence.**

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## Readiness for the connected explanation

What does `row["used"]` mean?

## This week's end goal

By the end, I can read a table schema and distinguish a numeric value from its text representation.

# 2. Notice the evidence

## PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use rows, columns, and data types to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

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My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

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## A closer explanation

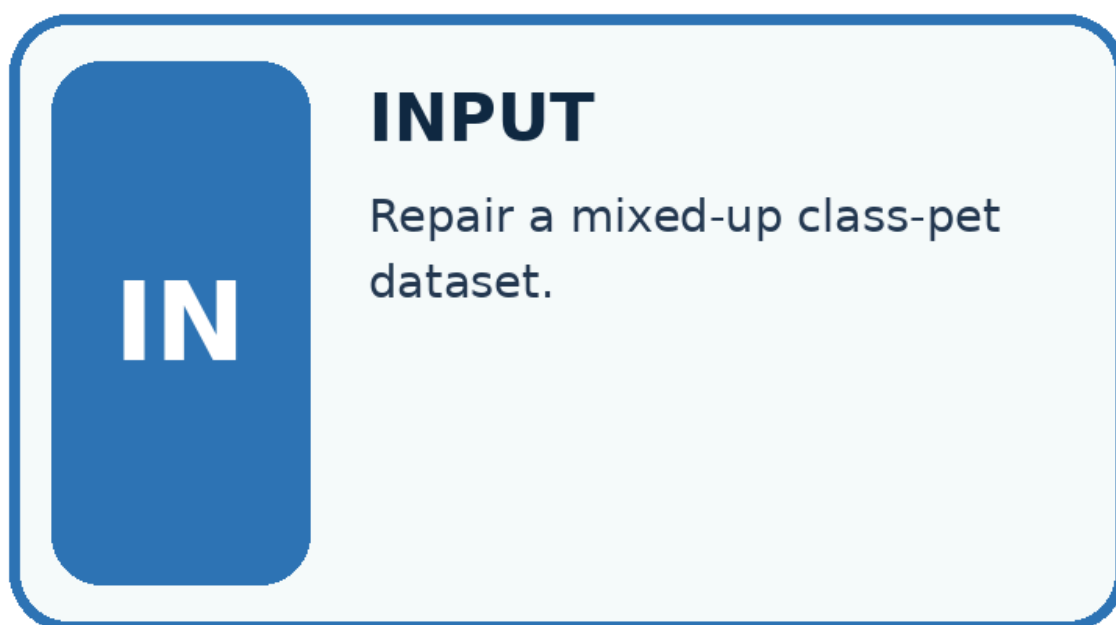
A fictional classroom logs the duration of each activity. Before comparing durations, define a schema: activity is nonempty text, minutes is an integer from 0 through 60, and each row describes one activity session. These decisions make data interpretable.

### 3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



## PROCESS

Structured tables preserve meaning through named fields and consistent types.



OUT

## OUTPUT

Repair a mixed-up class-pet dataset.

Which stage could fail even when the other two are correct? Explain:

**Write or draw your reasoning and evidence.**

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## A closer explanation

```
rows = [{"activity": "Reading", "minutes": 12},
        {"activity": "Sorting", "minutes": 0}]

for row in rows:
    valid = (type(row["activity"]) is str
             and len(row["activity"]) > 0
             and type(row["minutes"]) is int
             and 0 <= row["minutes"] <= 60)

    print(valid)
```

## Follow the evidence

### Step or case · Evidence · Meaning

**Step or case:** activity

**Evidence:** Nonempty text

**Meaning:** Names the session activity

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**Step or case:** minutes

**Evidence:** Integer 0 through 60

**Meaning:** Duration in minutes

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Read each row from left to right and connect the evidence to the stated rule.

The worked script prints True twice. `type(value)` is int checks the exact type here and avoids accepting a Boolean as a count. The chained comparison checks both boundaries. The example assumes both keys exist; next week handles missing fields. A zero duration is allowed by this written schema.

## Check a row in separate steps

A schema is a checklist for what a row must contain. First ask whether the activity name is text and is not empty. Next ask whether minutes is an integer. Then ask whether that integer lies between 0 and 60, including both ends. Keep these checks separate when explaining a rejection. The number 12 and the text "12" look similar on a page, but only the number meets this lesson's minutes rule.

## Connect the picture and the explanation

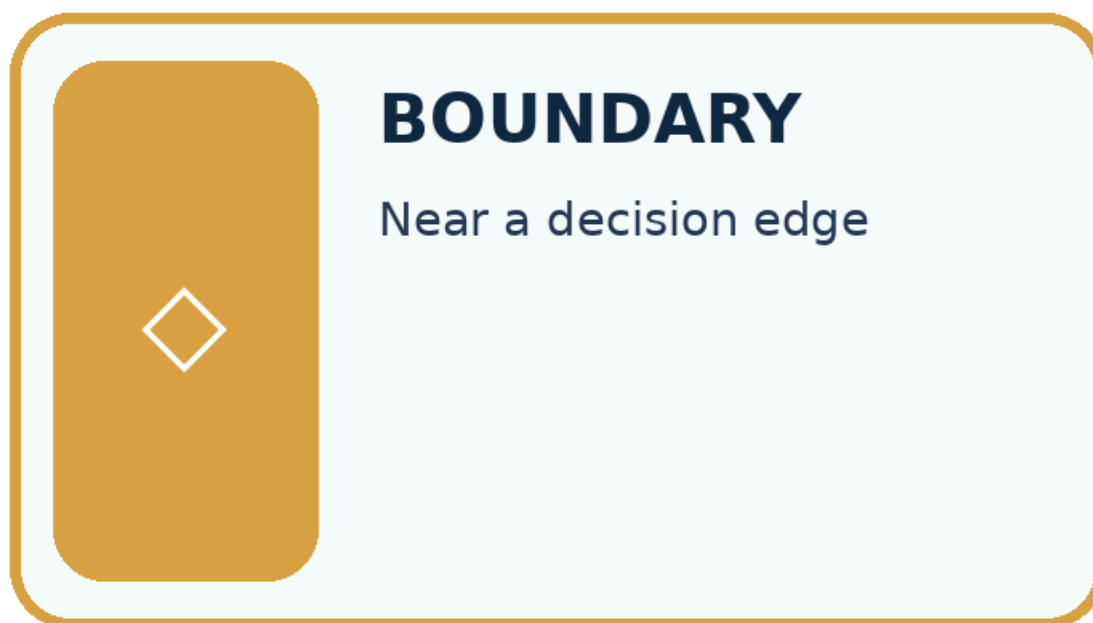
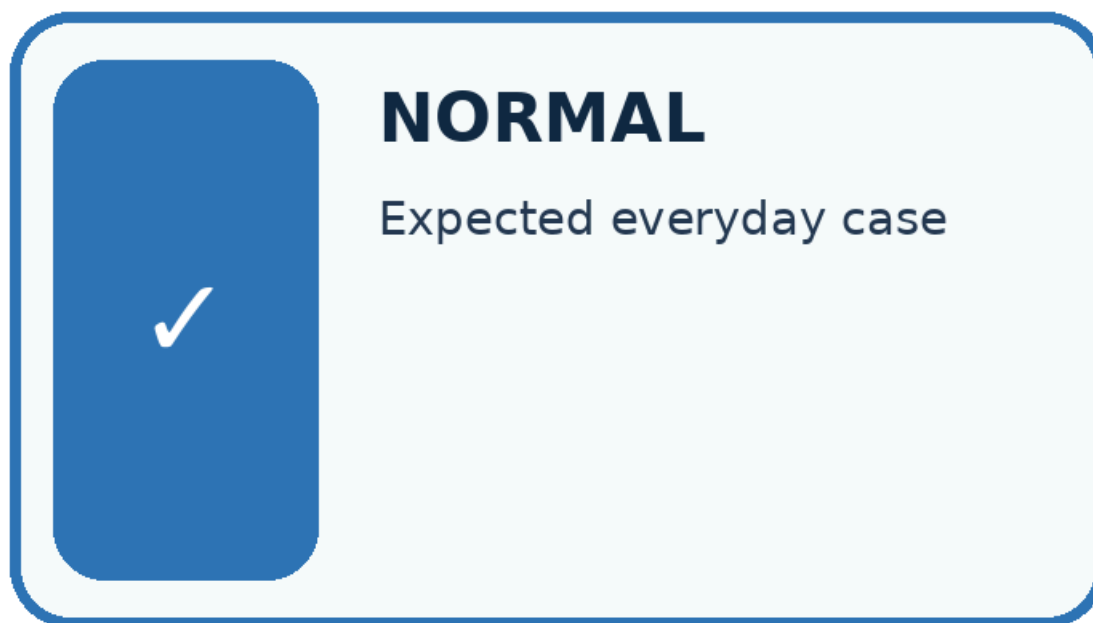
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

### 4. Compare the cases

#### CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.

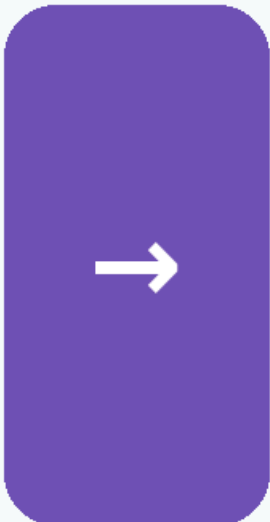


NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



## MISSING

Needed information absent



## TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

**Write or draw your reasoning and evidence.**

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### Change one thing in the pictured case

Change: Change B's pages from "90" to integer 90 using an approved repair.

Prediction: B now satisfies the declared schema.

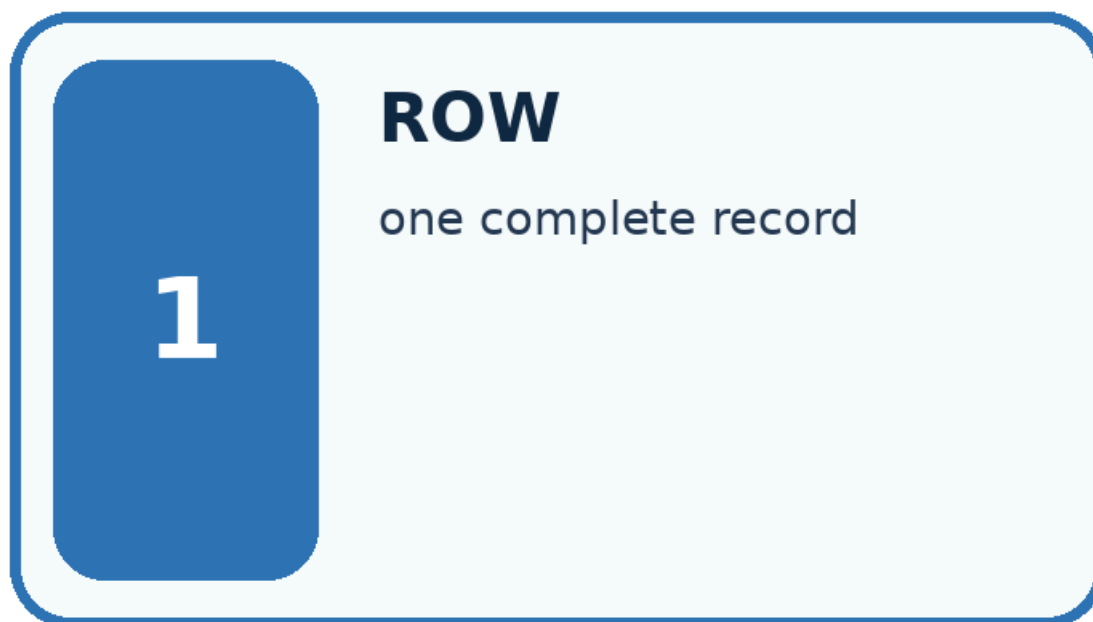
Reason: Its id is nonempty and illustrated is Boolean; the page-type defect is removed.

## 5. Words that unlock the idea

### VOCABULARY LAB

#### Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



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## **COLUMN**

one named field across records

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## **DATA TYPE**

kind of value allowed

4

# SCHEMA

the table's field rules

Word · My own example · Not this / boundary

**Word:** row

**My own example:** One record or example in a table.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** column

**My own example:** A table field containing the same kind of information across rows.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** data type

**My own example:** A category of values, such as number, text or Boolean, with defined operations.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** schema

**My own example:** A specification of required fields, types and structural rules for data.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Words used in the closer explanation**

## Word · Meaning

**Word:** Type

**Meaning:** A category such as integer or text

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**Word:** Unit

**Meaning:** The quantity scale attached to a number

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**Word:** Constraint

**Meaning:** A rule a valid record must satisfy

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## 6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

### WEEK 7 • WORKED EVIDENCE

- 1 Schema: title=text; pages=whole number; genre=category.
- 2 Bad record: pages='many' cannot be compared numerically.
- 3 Repair: verify the source, then enter 128—not a guess.
- 4 Consistent types let sorting and calculations work predictably.

1. Schema: title=text; pages=whole number; genre=category.
2. Bad record: pages='many' cannot be compared numerically.
3. Repair: verify the source, then enter 128—not a guess.
4. Consistent types let sorting and calculations work predictably.

Recalculate, retrace, or restate the reasoning in your own representation:

**Write or draw your reasoning and evidence.**

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CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

### A closer explanation

**Would minutes of 61, "12" and 12 each pass? Explain the type distinction.**

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### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Rows Columns and Data Types

A fictional approved book catalogue has three fields per row.

#### The facts we will use

Schema: id is nonempty text; pages is an integer 1–300; illustrated is Boolean True or False.

Rows A={id:"B7",pages:90,illustrated:True}; B={id:"B8",pages:"90",illustrated:False};  
C={id:"",pages:80,illustrated:True}.

## Rows Columns and Data Types

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Column contract

Field names identify values; order is not the only guide.



#### 2. Validate A

B7 is text, 90 is allowed integer, True is Boolean: valid.



#### 3. Validate B

Quoted "90" is text, so pages fails the integer contract.



#### 4. Validate C

Empty id fails even though its other two values have allowed types.

### CHANGE ONE THING

Change B's pages from "90" to integer 90 using an approved repair.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Column contract

**What happens and why:** Field names identify values; order is not the only guide.

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**Step:** 2. Validate A

**What happens and why:** B7 is text, 90 is allowed integer, True is Boolean: valid.

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### Step: 3. Validate B

**What happens and why:** Quoted "90" is text, so pages fails the integer contract.

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### Step: 4. Validate C

**What happens and why:** Empty id fails even though its other two values have allowed types.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

## Why this result follows

A schema makes row meanings and expected types explicit. The string "90" looks like a number when printed but does not meet this strict integer schema. A repair can be authorised later, but must not be assumed silently. Missing or empty identifiers also prevent tracing an item back to its source.

## What this example does not establish

Schema validity does not confirm that the book really has 90 pages or that its metadata is accurate.

Read the labelled picture: Which two rows originally fail, and do they fail for the same reason?

Change one thing: Change B's pages from "90" to integer 90 using an approved repair. Predict the result before checking the explanation. What remains the same?

# 7. Find and repair a mistake

## ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



### COMPARE • DIAGNOSE

Weak idea: A spreadsheet cell's color tells the computer its data type. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

**Evidence record:** normal case

**Expected:** works

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** boundary case

**Expected:** stated rule

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** missing input

**Expected:** safe response

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** fresh transfer

**Expected:** generalizes

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

Repair one cause. Why is this repair better than changing several things?

**Write or draw your reasoning and evidence.**

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## A closer explanation

A row can satisfy every schema rule and still be a wrong observation. Schema validation checks our stated

structure and constraints, not whether someone measured honestly or selected a representative sample.

**A row says Reading, 12 and passes. Does that establish the session really lasted 12 minutes?**

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## 8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



### TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence\_record PREDICT expected\_result RUN one normal case and one boundary case  
COMPARE actual\_result with expected\_result IF different: locate first cause; change one factor; RETEST  
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

**Write or draw your reasoning and evidence.**

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## A closer explanation

If one row records 120 seconds in the minutes field, a simple numeric total becomes misleading. Correct the source or explicitly convert and document it; renaming a column alone does not convert values.

## Practise with less help: Classify records

Use workbook W2 for the connected example “Rows columns and schemas”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Check your reasoning

Use the worked case for Rows columns and schemas. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

## Explain the picture

Which two rows originally fail, and do they fail for the same reason?

### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

### EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



### PREDICT → TEST → RECORD → EXPLAIN

Audit a 10-row book table. Circle mixed types, missing values, duplicated rows, and fields with unclear names; document each repair.

Audit a 10-row book table. Circle mixed types, missing values, duplicated rows, and fields with unclear names; document each repair.

**Test / input · Prediction · Actual · Evidence / calculation · Change or keep?**

**Test / input:** normal

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** boundary

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** missing

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** fresh transfer

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

What did the hardest test teach you that the normal test did not?

**Write or draw your reasoning and evidence.**

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**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



## MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

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# 11. Connect your project

## CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



## TRANSPARENT BOOK RECOMMENDER

Store one book per row and keep reader preferences in a separate record.

THIS WEEK'S CHECKPOINT Store one book per row and keep reader preferences in a separate record.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Publish the recommender's row schema and validation results before ranking books, preserving repair

# 12. Show what you know

## INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



**INDEPENDENT PROOF**

Repair a mixed-up class-pet dataset.

CHALLENGE Repair a mixed-up class-pet dataset.

1. My fresh example, trace, calculation, or design:

**Write or draw your reasoning and evidence.**

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2. My evidence that it works (include a boundary or missing-data case):

**Write or draw your reasoning and evidence.**

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3. One limitation, fairness concern, or situation needing human review:

**Write or draw your reasoning and evidence.**

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SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

## 13. Study a complete case

### Week 7 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

Fields are Title, Pages and Borrowed; a row contains River, ninety and yes.

#### The question

Assign intended types and a repair.

#### Worked explanation

Text, numeric and Boolean. Confirm ninety means 90, then convert and record the change; use a consistent Boolean representation.

#### Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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## Trace the supplied case visually

### A field has a meaning and a type

#### Evidence or stage · Value or result

**Evidence or stage:** Title

**Value or result:** Text → River

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**Evidence or stage:** Pages

**Value or result:** Number → 90 after checking source

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**Evidence or stage:** Borrowed

**Value or result:** Boolean → true or false

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Fields are Title, Pages and Borrowed; a row contains River, ninety and yes.

Assign intended types and a repair.

Text, numeric and Boolean. Confirm ninety means 90, then convert and record the change; use a consistent Boolean representation.

## 14. Try a new case independently

### Independent case: Design a schema independently

For a fictional shelf audit, define label and book\_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

#### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 7 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A table row is book\_id B7, pages 84, available true. Name the type of each field and explain why pages stored as eighty-four complicates calculation.

My answer, with a trace, calculation or supporting evidence

**Write or draw your reasoning and evidence.**

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### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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### Connect to your project

Store one book per row and keep reader preferences in a separate record.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

### Transfer practice from the original programme

A row contains title='Sky', pages='120', available='false'. Explain why treating nonempty text as true can misread availability. Propose explicit conversions and a rejected-input case.

**Explain the mechanism, evidence and one limit.**

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### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

## 15. Your lesson route

Start with this question: A fictional approved book catalogue has three fields per row.

**By the end, I can · How I will show it**

**By the end, I can:** Specify a table schema

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Check types and units as well as names

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Distinguish structural validity from truth

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Specify a table schema

**My deciding step, evidence and one remaining question**

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

## 17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional approved book catalogue has three fields per row.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Empty id fails even though its other two values have allowed types.

1. Column contract
2. Validate C
3. Validate B
4. Validate A

**My selected conclusion, deciding fact and explanation**

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Validate C Empty id fails even though its other two values have allowed types. A schema makes row meanings and expected types explicit. The string "90" looks like a number when printed but does not meet this strict integer schema. A repair can be authorised later, but must not be assumed silently. Missing or empty identifiers also prevent tracing an item back to its source.

## Practice 2: explain the deciding evidence

Changed-case practice: Change B's pages from "90" to integer 90 using an approved repair. Which conclusion is supported by the supplied facts?

1. B now satisfies the declared schema.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

**My selected conclusion, deciding fact and explanation**

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: B now satisfies the declared schema. Its id is nonempty and illustrated is Boolean; the page-type defect is removed.

## Repair a missing building block

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?. Return to this week and explain what you changed.

Review Trace Tables (ai-2/4). Goal: Trace variable values before and after each loop iteration. Short practice: Move print inside the original loop. What sequence is displayed?. Return to this week and explain what you changed.

Review Precision in Algorithms (ai-2/1). Goal: Translate a short paper calculation into a Python script. Short practice: If `print(total)` moves before `total=total+2`, what gets printed for the original input?. Return to this week and explain what you changed.

## Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

## Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.